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REGULAR PAPER

# An implicit layout-aware transformer for full-page end-to-end optical music recognition

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**Abstract**

The digital preservation and accessibility of musical manuscripts present a significant challenge in musical heritage conservation. Optical Music Recognition (OMR) is a key solution for automating the transcription of sheet music into machine-readable formats, typically relying on multi-stage processing due to the complexity of musical notation. Although the trend in OMR is to reduce the number of processing steps, the latest single-step approaches lose layout information from the original image, which is crucial for accurately interpreting musical documents in practical applications. This paper presents the Layout-Aware Sheet Music Transformer, a novel method based on a single-step transcription strategy that leverages transformer attention mechanisms to extract layout information without explicit annotations. By dynamically analyzing internal model representations during direct image-to-sequence transcription, our approach extracts nuanced structural insights without requiring manual annotations, eliminating manual layout annotations while offering deeper insights into document structures that more faithfully reflect the underlying musical meaning. Experiments in four well-known OMR datasets show that our proposal is both superior in transcription and competitive in layout extraction, making it a solid candidate for a new state of the art in the field.

**Keywords** Optical Music Recognition · Full page · Mensural notation · Layout analysis

## 1 Introduction

Across the world, countless musical manuscripts remain scattered, representing a vast cultural heritage in need of preservation [1, 2]. One way to safeguard these documents is through digitization, which involves scanning and converting them into image-based formats. While digitization preserves the visual representation of musical scores, true accessibility goes beyond mere image storage—it requires transcription into structured digital formats. However, the vast volume and diversity pose a significant challenge for manual transcription, being a costly and error-prone task. In order to overcome these challenges, Optical Music Recognition (OMR) emerged as the field dedicated to automating the transcription of musical scores [3]. By converting sheet music images into machine-readable formats, OMR enhances accessibility, facilitates computational music analysis, and significantly reduces the effort required by experts [4].

In real-world applications, an OMR system usually operates on entire sheet music pages for transcription. These systems, despite of being framed as *end-to-end*, typically implement multi-stage pipelines to get this output. The most common approach combines a Layout Analysis (LA) process that isolates the staves and a single-staff transcriptor. These approaches are typically implemented through deep learning [5–7]. While dividing the task into multiple steps provides some benefits, such as simplifying the individual processes, it can also introduce errors that propagate to subsequent stages. This fact requires an intermediate review by the user to fix possible errors. This is an aspect that the literature typically overlooks. In addition, these pipelines require obtaining a fine-grained labeled training set for each task, hardening their practical implementation.

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Recent research in OMR has delved into producing single-step systems that are able to provide the content of the score without any LA. These solutions have proven effective in many use-cases, even in documents that were considered out of reach, such as pianoform scores [8, 9]. Despite their benefits, the resulting digital transcription misses information about the original layout of the score, which severely hinders the faithful reproduction of the document, which is essential for practical OMR applications.

Taking advantage of the simplicity of these end-to-end full-page methodologies, we propose the Layout-aware Sheet Music Transformer (LA-SMT), a novel end-to-end OMR method—based on the recent Sheet Music Transformer (SMT) [9, 10]—that is able to both transcribe and give layout-image information from an input full-page score image without requiring explicit graphic annotations. Our approach leverages the internal representations of the Transformer attention mechanisms during the direct image-to-sequence transcription process to infer the structural organization of the input document. Similarly to this idea, several works can be found in the literature, even applied to other fields, as the case of [11] or [12]. This methodology aims to reveal the document structure intrinsically connected to its musical content and output a complete digital version that reflects the complete information of the original manuscript. In this paper, we conduct a study in the context of full-page recognition, in contrast to the single-staff focus of the literature. Our experiments show that our method outperforms traditional approaches in transcription performance and is competitive in staff detection against well-established object detectors. This analysis introduces a new state-of-the-art method in the field of OMR, improving upon previously achieved results while retrieving document structure from the inner features of the model.

## 2 Related work

Optical Music Recognition (OMR) has traditionally followed a sequential pipeline, involving preprocessing steps like binarization and staff-line removal [13, 14], followed by symbol recognition [15, 16] and final encoding into symbolic formats [17]. Each step in this pipeline depends on the accuracy of previous processes, meaning that errors in the early stages can propagate and negatively impact the final result.

Recent advances in deep learning have driven a trend towards reducing the number of processing steps in OMR systems, keeping the performance and generalization capabilities [18]. This strategy alleviates the need for specially labeled data for each of the tasks involved. Most of the literature focuses on single-staff scenarios. They are mainly based on the Convolutional Recurrent Neural Networks (CRNNs) combined with Connectionist Temporal Classification (CTC), which captures both spatial and sequential dependencies in music notation. This architecture has been widely established in the community for both printed and handwritten scores [19–24]. Other approaches are also found, such as the use of next-symbol prediction [25] or Graph Neural Networks [26].

All of these works assume that the input sources are single-staff images that have been extracted in a previous step. This requires pipeline-based solutions for processing full-page music scores.

This typically has been addressed through object detection, where the bounding boxes of single staves are extracted for later processing [27, 28]. The work of Castellanos et al. explored several general-purpose object detectors for this task [29]. The YOLO architecture has also gained popularity in the field [43]. Despite their effectiveness, these approaches heavily rely on large amounts of annotated training data, requiring detailed bounding box annotations for staves.

As introduced before, a recent trend is to treat OMR as a single end-to-end process that operates at the full-page level. The first works adapted the CRNN via an unfolding mechanism, inherited from handwritten recognition [8]. Nevertheless, inspired by the recent progress in this same area [30, 31], the SMT was proposed as the first complete end-to-end approach for full-page OMR, as it is capable of processing an entire sheet music image to produce its encoded transcription in a single step [9, 10]. In fact, this approach is demonstrated in pianoform music scores, which are the most complex documents in the literature. Although this model has marked a significant advance in the field, it does not provide information about the structure of the image, losing valuable information for practical OMR. This paper specifically aims to fill this gap in the OMR literature by retrieving structural information from this single-step method.

Several works in other domains have explored the use of attention mechanisms to extract additional information. For instance, [32] and [33] analyzed attention patterns in Natural Language Processing tasks to better understand how contextual information is propagated across layers, with implications for alignment and interpretability. Another example is that by [34], who leveraged attention not only for transcription but also as a means to guide detection and localization of text in the field of scene text recognition. These approaches show that attention maps can provide useful spatial data, although their application was not demonstrated in the field of OMR. This paper addresses this gap in the literature by contributing a methodology for an implicit layout-aware full-page OMR.

# 3 Layout-aware sheet music transformer

In this section, we propose the LA-SMT. This is the first comprehensive end-to-end approach that integrates both transcription and structural information extraction from documents in only one step. The input is an image of a scanned sheet music page, while the output consists of its transcription along with the corresponding layout analysis.

## 3.1 Sheet music transformer

The method employs an end-to-end neural network architecture for full-page music score transcription, based on the SMT framework [10]. The model comprises two primary components: an encoder that extracts high-level feature representations and a decoder that generates content autoregressively. Let $x$ be a full-page music score image normalized to [0, 1], where $x \in \mathbb{R}^{c\times h\times w}$ ($h$ and $w$ are pixel dimensions, $c$ is the number of channels). The encoder transforms this input into a feature map $x'_e \in \mathbb{R}^{h_e\times w_e\times c_e}$, with spatial dimensions $h_e = \frac{h}{r_h}$ and $w_e = \frac{w}{r_w}$, reduced by pooling operations. The Transformer decoder [35] then generates output tokens according to a vocabulary, represented as $\Sigma$, by maximizing the conditional probability:

$$\hat{y}_t = \arg\max_{y_t\in\Sigma} P(y_t \mid x'_e, (\hat{y}_0, \hat{y}_1, \dots, \hat{y}_{t-1})) \tag{1}$$

This iterative process continues until an end-of-sequence token is predicted. Leveraging autoregressive generation and Transformer attention mechanisms, the SMT transcribes the entire music score without explicit segmentation.

## 3.2 Implicit layout analysis extraction

To describe the process of structural information extraction from the image, henceforth referred to as Implicit Layout Analysis (ILA), it is essential to understand how the attention mechanism operates within the decoder. The Transformer decoder architecture is composed of two main blocks: self-attention and cross-attention. In self-attention, the language model generates a coherent and contextualized representation of the transcription, enabling accurate outputs. The cross-attention correlates the information of the transcription with the image. In an analogous comparison, it operates similarly to the human reading eye, which selects specific symbols for precise transcription from the feature map $x'_e$. This process produces a sparse attention matrix, where weights concentrate on specific areas of the image. Collectively, these matrices across inference steps approximate the relevant layout information of the score. In this paper, we present a method that leverages this feature to extract the layout regions from the mapped attention matrices of the SMT during inference. A general overview of the process can be found in Fig. 1, while Algorithm 1 represents the proposed scheme for retrieving document structure from the model.

**Algorithm 1** Implicit Layout Analysis (ILA)

1: **function** ILA($\hat{\mathbf{y}}$, $\mathcal{M}$, $n$)
2: $\mathcal{R} \leftarrow \emptyset$ ▷ Initialize regions set
3: $M_r \leftarrow \emptyset$ ▷ Initialize accumulated matrices
4: **for** $i \leftarrow 1$ **to** $n$ **do**
5: **if** region-split-criterion($\hat{\mathbf{y}}_i$, $\mathcal{M}_i$) **then**
6: $M' \leftarrow$ combine-matrices($M_r$)
7: $M' \leftarrow$ rescale($M'$, $r_h$, $r_w$)
8: $M' \leftarrow$ dilate($M'$, $(d_h, d_w)$, $n$)
9: $C \leftarrow$ detect-contours($M_{dilated}$)
10: $P \leftarrow$ convex-hull($C$)
11: $(x, y), h, w \leftarrow$ dimensions(P)
12: $\mathcal{R} \leftarrow \mathcal{R} \cup \{(x, y, h, w)\}$
13: $M_r \leftarrow \emptyset$ ▷ Reset accumulated matrices
14: **else**
15: $M_{binary} \leftarrow$ binarize($\mathcal{M}_i$, $u$)
16: $M_r \leftarrow M_r \cup \{M_{binary}\}$
17: **end if**
18: **end for**
19: **return** $\mathcal{R}$
20: **end function**

Let $\hat{\mathbf{y}}$ represent the list of predicted tokens from the SMT, and let $\mathcal{M}$ be the corresponding list of $n$ cross-attention matrices for each token (*line 1*). The algorithm initializes an empty set of regions $\mathcal{R}$ and a temporary set of matrices $M_r$ to accumulate attention maps (*lines 2–3*). The main loop (*line 4*) iterates over each $i$-th predicted token and its corresponding attention matrix $M_i$. For each token $i$, the algorithm evaluates a region-splitting criterion to determine whether the current staff has been fully processed (*line 5*). This criterion is used to segment the sequence into meaningful layout regions, see Section 3.2.1 for details. If the criterion is met, the algorithm proceeds to: (i) combine the accumulated matrices $M_r$ into a single one $M'$ (*line 6*); (ii) rescale it to the original image dimensions using the parameters $r_h$ and $r_w$, which represent the horizontal and vertical scaling ratios to be applied (*line 7*), (iii) apply dilation with a kernel of size $d_h \times d_w$ and $n$ iterations (whose specified values are described in Section 4.4) to expand symbol activations (*line 8*); (iv) detect contours and compute the convex hull of the detected region (*lines 9–10*); (v) extract the position $(x, y)$, height $h$, and width $w$ of the resulting polygon (*line 11*); and (vi) add the resulting bounding box to the set of regions $\mathcal{R}$ (*line 12*).

After that, the matrix buffer $M_r$ is cleared for the next region detection (*line 13*). If the region-splitting condition is not met, the current attention matrix $\mathcal{M}_i$ is binarized using a fixed-valued threshold $u$ (*line 15*), and added to the buffer $M_r$ (*line 16*), allowing accumulation over consecutive tokens. Finally, the algorithm returns the complete set of detected layout regions $\mathcal{R}$ (*line 19*).

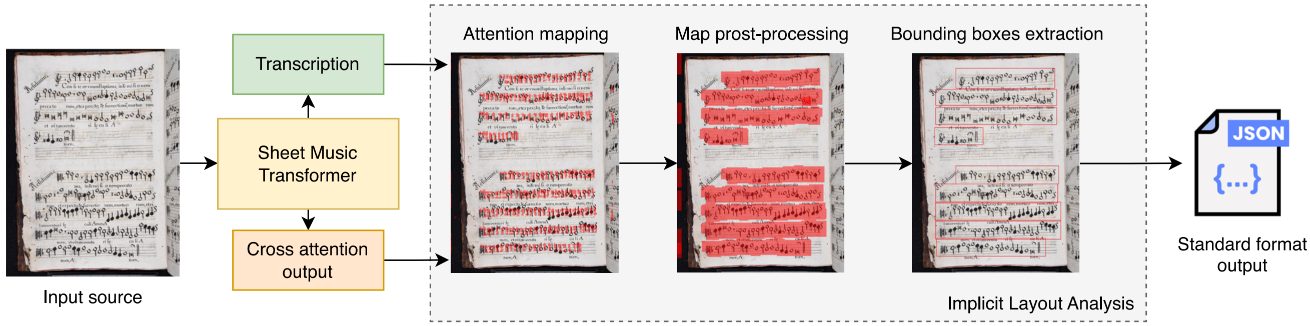


**Fig. 1** Implicit Layout Analysis proposed system to output a complete digital document of a given input full-page music score

#### 3.2.1 Region splitting criteria

The most critical component of the Implicit Layout Analysis (ILA) algorithm is the region splitting criterion, which automatically determines which parts of the prediction constitute a region. In this paper, we propose three different splitting criteria. These methods rely exclusively on the predicted content and the cross-attention matrices generated by the SMT. Since different variants are proposed, we will henceforth refer to our method as LA-SMT$^{\lambda}$, where lambda denotes the criterion being used.

***Language***
Music scores share a common graphical pattern: all staves begin with a clef symbol. The first criterion leverages this characteristic by defining regions based on the predicted clefs. Specifically, the set $M_r$ is populated as long as the predicted token $\hat{y}_i$ is not a clef symbol. When a clef is detected, the algorithm processes the accumulated region and initiates a new one, continuing this way until another clef is found. This criterion is the most straightforward, as it only requires direct token comparison to determine whether a new region should be created. This proposal will be referred to as LA-SMT$^{\text{Lang}}$.

***Attention distances***
The language-based criterion, although straightforward, implicitly assumes that each staff in a music score contains only one clef. However, some scores may include inner-clef changes, as is the case in certain pianoform works [4]. Moreover, this criterion depends on the model's ability to accurately transcribe all clefs present in the document, which is not always guaranteed.

In contrast, the attention distance criterion is graphic-based, relying on vertical distances between attention matrices. As described in Sect. 3.2, the general algorithm binarizes and stores the individual attention matrices corresponding to each predicted token. These matrices approximate the position of each predicted symbol and are stored in sequential reading order. The criterion computes the approximate positions of the symbols—using the same method as in Algorithm 1—and measures the distance between them. If the gap between two consecutive symbols exceeds a predefined threshold, it is assumed that the reading order has shifted to a new region. In this way, Implicit Layout Analysis (ILA) can be implemented in a language-agnostic manner, ensuring that transcription errors do not affect the accuracy of layout extraction. This proposal will be referred to as LA-SMT$^{\text{AD}}$.

***Layout tokens***
Despite the theoretical advantages offered by the attention distance criterion, it can be computationally inefficient, as it requires computing a bounding box for each symbol in order to measure the leaps between predictions. Moreover, this method depends on the precision of the SMT in localizing symbols via attention—a capability that is not explicitly enforced during training.

The third proposed criterion for Implicit Layout Analysis (ILA) involves the inclusion of layout tokens in the output ground truth, following an approach similar to that used in Handwritten Document Recognition [30, 31]. Layout tokens act as special markers—analogous to those in markup languages—that enclose sequences of symbols belonging to a specific region of the score. During training, the model learns to predict these boundary tokens alongside the content tokens, establishing a link between the image layout and the transcription without requiring explicit segmentation.

This approach, referred to as LA-SMT$^{\text{LT}}$, offers both computational efficiency and robustness to transcription errors, as is demonstrated in our experiments. Additionally, it provides an explicit representation of region boundaries, which can be leveraged in downstream applications such as score rendering or interactive editing without significantly increasing vocabulary complexity.

## 4 Experimental setup

This section presents the experimental setup, including details about the corpora used, the evaluation metrics considered, and the baseline methods against which our approach

**Table 1** Summary of the corpora considered for our experiments with the number of staves, symbols, and pages, and their engraving type, whose values can be 'Ty.' for typeset and 'Hw.' for handwritten

| Property | Corpus | | | |
|---|---|---|---|---|
| | MOT | SEILS | CAP | GUAT |
| *Num. pages* | 281 | 150 | 123 | 355 |
| *Num. staves* | 1 847 | 1 100 | 704 | 3 263 |
| *Total symbols* | 41 050 | 32 556 | 12 408 | 72 504 |
| *Unique symbols* | 228 | 183 | 321 | 315 |
| *Engraving* | Ty. | Ty. | Hw. | Hw. |

is compared. For the sake of reproducibility, the source code is publicly available.[1] The source code of the experiments is available at https://github.com/antoniorv6/smt-implicit-la

### 4.1 Corpora

The experimental evaluation is conducted on four distinct corpora of mensural music notation, which represent the main benchmarks for developing state-of-the-art end-to-end full-page OMR systems. Table 1 summarizes the key statistics of each corpus, including the number of pages, staves, and symbols, as well as the engraving type. A brief description of each corpus follows:

- MOTTECTA [36]: a collection of 281 typeset pages from mensural music books housed in the "Biblioteca Digital Hispánica", dating back to the 17th century. An example of a page from this corpus is depicted in Fig. 2a.
- SEILS [37]: a set of 150 printed pages from an anthology of 16th-century Italian madrigals in mensural notation, particularly *Second Edition Il Lauro Secco*. Figure 2b showcases an example page from this corpus.
- CAPITAN [19]: a collection of 96 pages of liturgical music in mensural notation, dating back to the 17th century. For the purposes of the experiments, as they were conducted on single-page data, six double-page manuscripts were excluded, resulting in a final dataset of 90 handwritten pages. An example of this corpus is shown in Fig. 2c.
- GUATEMALA [38]: a set of 355 handwritten pages from a monophonic choir book, part of a larger collection preserved at the "Archivo Histórico Arquidiocesano de Guatemala", composed from the 16th to the 18th centuries. Fig. 2d depicts an example page from this corpus.

Each corpus was annotated using the *agnostic* format, a structured representation that abstracts musical content from any specific notation style. In this format, each note is encoded in the form $<\texttt{shape}>:<\texttt{height}>$, where $<\texttt{shape}>$ specifies the notehead shape, while $<\texttt{height}>$ refers to its relative pitch position within the staff. This encoding ensures that the transcription remains independent of the original graphical conventions while preserving essential musical semantics [39].

To evaluate our methodology and draw generalizable conclusions, we employed a 5-fold cross-validation scheme across all four datasets. In each fold, the data was randomly split into 20% for testing, 20% for validation, and the remaining 60% for training.

### 4.2 Metrics

To assess the effectiveness of our proposal, we employ distinct evaluation metrics for LA and OMR, chosen based on their established use in the literature. Each metric provides complementary insights into different dimensions of model performance, allowing for a comprehensive evaluation of both structural layout extraction and transcription accuracy.

***Layout Analysis metrics***
We evaluated the layout extraction process using standard object detection and segmentation metrics commonly adopted in document analysis. The first is the Intersection over Union (IoU), which measures the overlap between the predicted and ground-truth bounding boxes. It is defined as:

$$\mathrm{IoU} = \frac{|B_p \cap B_t|}{|B_p \cup B_t|} \tag{2}$$

where $B_p$ denotes the predicted bounding box and $B_t$ the ground truth.

The second metric is the mean Average Precision at COCO thresholds (mAP@COCO), also referred to as mAP @[0.50:0.95], which is the primary evaluation metric in the COCO object detection challenge [40]. This metric assesses detection performance across a range of Intersection over Union (IoU) thresholds, offering a more rigorous and comprehensive evaluation. Specifically, it computes the average precision at ten different Intersection over Union (IoU) thresholds, ranging from 0.50 to 0.95 in increments of 0.05.

***Transcription quality metric***
In the context of OMR, the Symbol Error Rate (Symbol Error Rate (SER)) is a widely adopted metric for assessing transcription quality. It is an adaptation of the Character Error Rate (CER), commonly used in text transcription, to the musical domain. This metric is grounded in the concept of edit distance, quantifying the number of operations required to transform the model prediction into the ground truth. Unlike the metrics defined for LA, where higher values typically indicate better performance, a lower (Symbol Error

[1] The source code will be publicly available upon acceptance for publication.

(a) MOTTECTA (b) SEILS (c) CAPITAN (d) GUATEMALA

**Fig. 2** Examples of pages from the corpora considered for the experiments. MOTTECTA and SEILS correspond to printed or typeset documents, whereas CAPITAN and GUATEMALA include handwritten manuscripts. Characteristic can be found in Table 1. All the images were annotated by the Muret tool

Rate (SER)) corresponds to a more accurate transcription. In all cases, we have considered SER at the page level.

### 4.3 Baseline approaches

In the OMR literature, obtaining document layout information typically requires a multi-stage pipeline, which usually begins with the application of object detection or segmentation methods to localize the structural components of the document. After the LA stage, the extracted regions are individually processed by a transcription stage, typically a CRNN+CTC. Both stages are described below. Concerning the post-processing to extract the bounding boxes, after preliminary experiments, we determined a $d_h \times d_w$-kernel of $5 \times 5$ and $n = 2$ iterations in the dilation step. Note that this is approximately equivalent to a single dilation with a larger kernel (e.g., $9 \times 9$ or $11 \times 11$). However, using smaller kernels with multiple iterations may provide more limited expansion and reduce the risk of unintentionally merging nearby symbols—an important consideration given the dense and fine-grained structure of musical notation. Although this choice does not significantly affect the current task (staff retrieval), it could have a negative impact on other OMR tasks, such as precise symbol positioning.

#### 4.3.1 Layout Analysis pipeline

Regarding LA, several representative methods from the state of the art have been selected. These methods aim to detect and isolate staff regions from full-page music scores for subsequent transcription. The first method is Selectional Auto-encoder (SAE) [41], a two-step approach based on the U-Net architecture, which provides a confidence map indicating the class membership of each pixel. This map is then processed using connected component analysis to extract the bounding boxes of the staves. The second baseline method is RetinaNet [42], a general-purpose object detector designed to achieve high accuracy while maintaining computational efficiency. RetinaNet follows a single-stage approach, making it faster while still achieving high accuracy. It was selected as one of the most proper methods based on previous work [29]. Lastly, we include the YOLO architecture in our experimentation, a model that provides a competitive balance between detection accuracy and computational cost. Recent research revealed that YOLO-based architectures are a potential solution for LA [43]. In this case, we considered a recent version of the architecture, specifically the YOLOv11-s.

While Selectional Auto-encoder (SAE) processes the images after resizing them to $512 \times 512$, the rest of the models consider an input size of $640 \times 640$, as is commonly used in the literature. The models were trained with an early stopping technique of five epochs. It is important to note that all these models require a dedicated training set specifically annotated for staff detection—a prerequisite that is not necessary in our proposed approach.

#### 4.3.2 Transcription pipeline

Regarding the transcription pipeline, we consider the most common approach in the literature, which uses an end-to-end strategy based on a CRNN combined with CTC. This architecture integrates a CNN for feature extraction and an RNN for sequence modeling, allowing the network to capture both spatial and temporal dependencies in the musical notation. The CTC loss function enables the model to learn alignments between input features and the corresponding transcription without requiring explicit frame-level annotations. During inference, the model produces a probability posteriorgram, which is then converted into a sequence using a greedy decoding approach. The CRNN-CTC framework has been widely adopted in OMR because of its high performance and flexibility, even without requiring pre-training to obtain competitive results [7, 19, 36].

### 4.4 SMT configuration

Concerning the architecture employed in our proposal, we base it on SMT [10], which is a transformer-based model capable of processing a full-page image and obtaining its complete transcription in a single step. Despite its great learning capacity, one of its main limitations is the need for large amounts of training data. To tackle this limitation, we have considered pretraining the model with synthetic data from the PRIMENS dataset, which is an adaptation of the well-known PrIMuS dataset for mensural music transcription [22, 36]. Our model includes two ConvNeXT blocks as the encoder [44]—which reduce the input image by a factor of 8— and eight decoder layers with a hidden size of 256. It also incorporates four attention heads per layer, which are key for retrieving layout information.

### 4.5 Other end-to-end approaches

To provide a more comprehensive evaluation of our methodology for end-to-end applications, we compare the LA-SMT model against two additional transformer-based full-page transcription networks. The first is the Document Attention Network (DAN), currently regarded as the state-of-the-art for full-page handwritten text recognition [31]. This model employs the same decoder architecture as the SMt, but integrates a lightweight, fully-connected convolutional neural network as its encoder. This encoder reduces the input image by a factor of 4 in width and 16 in height [45]. We also evaluate the TrOMR network, another transformer-based architecture designed for the transcription of polyphonic music [46]. This model utilizes a customized ResNet as its feature extractor—a widely recognized architecture in computer vision. In particular, the TrOMR encoder downsamples the input image by a factor of 32 in both height and width.

## 5 Results & discussion

In this section, we report the performance reported by the proposed methods for end-to-end full-page OMR and compare their capabilities in both printed and handwritten scenarios. We also conduct a deeper qualitative analysis to assess the practical implications of these performance metrics.

### 5.1 Results on printed scores

Table 2 reports the performance across the two printed datasets: MOTTECTA and SEILS.

On the MOTTECTA dataset, the baseline transcription—which uses manually annotated staves as the test set—achieves a SER of 7.1%. Among the state-of-the-art approaches, the pipeline employing YOLO yields the best results, with SER = 6.7%, IoU = 88.5%, and a mAP@COCO = 86.5%. The SEA performed comparably well with a relative drop of $\approx$ 10% SER. RetinaNet lagged significantly, reporting a drop in both transcription and LA metrics. In the case of SEILS, both YOLO and the SAE perform similarly in transcription, although YOLO demonstrates superior detection capabilities, supported by a 3% and 19% improvement over the SAE.

The three variants of LA-SMT also yield strong performance in both datasets. In the case of MOTTECTA, the LA-SMT$^{LT}$ reports the best transcription performance with a 6.1% SER, which improves the YOLO pipeline over an 8%. This is also maintained over the SEILS dataset, where our proposals remarkably outperform the state of the art, with an overall improvement of 58%. However, our approach, although strong, did not yield better performance in the LA task.

Compared to the other two transformer-based architectures (TrOMR and DAN), our model achieves consistently higher performance in both transcription and layout analysis metrics. While transcription results are relatively competitive across models, the differences in layout analysis are more pronounced. For instance, TrOMR$^{LT}$ achieved its best IoU score on MOTTECTA with 65.3%, and 63.5% on SEILS. Similarly, DAN$^{LT}$, the best-performing variant of DAN, achieved 72.4% and 71.2% IoU on the same datasets. In contrast, our proposed model, LA-SMT$^{LT}$, achieved 80.9% IoU under the same stopping criteria. This performance gap may be attributed to the aggressive image reduction factors used in the other approaches—4 $\times$ 16 in TrOMR and 32 $\times$ 32 in DAN—which lead to a substantial loss of spatial detail and distortion of the original aspect ratio. Our approach, in contrast, uses a more moderate reduction factor of 8 $\times$ 8, preserving higher-resolution visual features and maintaining the aspect ratio. This preservation is particularly important for accurate layout analysis and transcription in complex historical documents. Similar trends can be observed when comparing models using the mAP metric, further supporting the superiority of our approach with the three stopping criteria considered.

In LA, our approach gets the best IoU in SEILS, while it is the second-best result in MOTTECTA, making it a robust option. However, YOLO seems to be superior in the mAP@COCO benchmark.

**Table 2** Average results among the five folds (%) for layout analysis (IoU and mAP@COCO) and transcription SER of the printed datasets. The figures in bold represent the best result for each metric in each of the two datasets for the typeset scenario, whereas underlined figures highlight the second-best result. Transcription results of SOTA were obtained by a CRNN+CTC applied to the retrieved staves by the object detection approaches. The same transcription model was employed with the ground-truth staves as a reference, represented by Gold LA

| | Method | Metrics (%) | | |
|---|---|---|---|---|
| | | IoU | mAP@COCO | SER per page |
| | | MOTTECTA | | |
| SOTA | `Gold LA` | - | - | 7.1 ± 0.5 |
| | `SAE` | 81.4 ± 1.1 | 60.6 ± 2.2 | 7.4 ± 0.6 |
| | `RetinaNet` | 73.2 ± 0.7 | 50.1 ± 1.2 | 21.8 ± 2.2 |
| | `YOLO` | **88.5 ± 0.6** | **81.7 ± 1.2** | <u>6.7 ± 0.4</u> |
| Other | TrOMR$^{LT}$ | 65.3 ± 4.1 | 51.8 ± 2.7 | 10.9 ± 0.8 |
| | TrOMR$^{AD}$ | 63.8 ± 4.4 | 49.6 ± 3.0 | 11.3 ± 0.9 |
| | TrOMR$^{Lang}$ | 64.1 ± 3.9 | 50.7 ± 2.5 | 11.3 ± 0.9 |
| | DAN$^{LT}$ | 72.4 ± 3.7 | 55.1 ± 3.8 | 6.9 ± 0.9 |
| | DAN$^{AD}$ | 70.1 ± 5.0 | 50.7 ± 3.1 | 7.2 ± 1.0 |
| | DAN$^{Lang}$ | 71.2 ± 3.4 | 53.6 ± 2.5 | 7.2 ± 1.0 |
| Ours | LA-SMT$^{LT}$ | 80.6 ± 6.9 | <u>67.4 ± 4.0</u> | **6.1 ± 0.8** |
| | LA-SMT$^{AD}$ | 79.0 ± 5.0 | 53.9 ± 5.8 | 7.7 ± 0.9 |
| | LA-SMT$^{Lang}$ | <u>81.9 ± 6.8</u> | 67.3 ± 1.3 | 7.7 ± 0.9 |
| | | SEILS | | |
| SOTA | `Gold LA` | - | - | 10.3 ± 4.1 |
| | `SAE` | 84.5 ± 2.6 | 70.8 ± 3.1 | 10.2 ± 3.0 |
| | `RetinaNet` | 80.1 ± 2.5 | 50.1 ± 10.0 | 39.8 ± 10.1 |
| | `YOLO` | <u>87.5 ± 4.0</u> | **84.3 ± 2.5** | 10.2 ± 4.2 |
| Other | TrOMR$^{LT}$ | 63.5 ± 3.8 | 50.9 ± 3.1 | 10.6 ± 1.0 |
| | TrOMR$^{AD}$ | 62.0 ± 4.1 | 48.7 ± 2.9 | 11.5 ± 0.9 |
| | TrOMR$^{Lang}$ | 62.7 ± 3.7 | 49.9 ± 2.8 | 10.4 ± 1.1 |
| | DAN$^{LT}$ | 71.2 ± 3.7 | 55.3 ± 2.9 | 5.1 ± 1.1 |
| | DAN$^{AD}$ | 68.7 ± 3.8 | 50.2 ± 3.3 | 5.4 ± 1.0 |
| | DAN$^{Lang}$ | 69.3 ± 3.5 | 52.6 ± 2.5 | 5.4 ± 1.0 |
| Ours | LA-SMT$^{LT}$ | 80.9 ± 6.0 | 62.9 ± 2.2 | <u>4.7 ± 2.6</u> |
| | LA-SMT$^{AD}$ | **88.1 ± 2.9** | 61.6 ± 2.8 | **4.2 ± 1.4** |
| | LA-SMT$^{Lang}$ | 86.7 ± 4.9 | <u>72.0 ± 1.3</u> | **4.2 ± 1.4** |

## 5.2 Results on handwritten scores

Table 3 shows the reported performance of the evaluated methods in CAPITAN and GUATEMALA.

In the case of CAPITAN, the base transcription system yields a 14.5% SER, while in GUATEMALA reports a 4%. This shows that, although both datasets are handwritten, CAPITAN is more challenging. This is mainly because of its smaller size, as reported in Table 1.

The LA task poses a significant challenge for RetinaNet, which delivers the worst results among the evaluated models, with a notable gap among its competitors. In contrast, the YOLO-based pipeline achieves impressive detection performance, with scores exceeding 90% on GUATEMALA and 80% on CAPITAN.

However, this pipeline is notably constrained by the transcription network. Specifically, in CAPITAN, we observe a drop in transcription performance of up to 11%, whereas in GUATEMALA, performance remains stable.

The LA-SMT, in contrast, still shows superior transcription performance in both sources, independently of the ILA stop criterion. In this case, the proposal improves the YOLO pipeline by 20% in GUATEMALA and 18% in CAPITAN. However, they do not excel in the detection performance metrics, where all IoUs are below the 80%, in contrast to the remarkable results of YOLO, which is over 90%. It is worth highlighting that the proposed models do not rely at any point on annotated reference data indicating the position of the staves. Despite this, they achieve competitive results, improving some SOTA cases and demonstrating the robust-

**Table 3** Average results among the five folds (%) for layout analysis (IoU and mAP@COCO) and transcription SER of the handwritten datasets. The figures in bold represent the best result for each metric in each of the two datasets for the manuscript scenario, whereas the underlined figures highlight the second-best result. Transcription results of SOTA were obtained by a CRNN+CTC applied to the retrieved staves by the object detection approaches. The same transcription model was employed with the ground-truth staves as a reference, represented by Gold LA

| | Method | Metrics (%) | | |
|---|---|---|---|---|
| | | IoU | mAP@COCO | SER per page |
| | | CAPITAN | | |
| SOTA | `Gold LA` | – | – | 14.5 ± 1.2 |
| | `SAE` | <u>87.1 ± 1.5</u> | <u>74.0 ± 3.2</u> | 19.2 ± 6.9 |
| | `RetinaNet` | 27.1 ± 21.3 | 7.9 ± 9.9 | 88.6 ± 15.5 |
| | `YOLO` | **91.0 ± 0.4** | **86.5 ± 2.2** | 16.2 ± 4.0 |
| Other | TrOMR$^{LT}$ | 58.4 ± 5.1 | 43.2 ± 3.0 | 19.5 ± 1.4 |
| | TrOMR$^{AD}$ | 56.1 ± 4.6 | 41.5 ± 2.8 | 18.6 ± 1.2 |
| | TrOMR$^{Lang}$ | 56.8 ± 4.9 | 42.3 ± 2.6 | 18.6 ± 1.2 |
| | DAN$^{LT}$ | 63.8 ± 4.5 | 46.0 ± 2.8 | 13.4 ± 1.7 |
| | DAN$^{AD}$ | 61.2 ± 3.6 | 42.5 ± 3.1 | 14.7 ± 1.9 |
| | DAN$^{Lang}$ | 60.5 ± 4.2 | 45.2 ± 2.0 | 14.7 ± 1.6 |
| Ours | LA-SMT$^{LT}$ | 70.5 ± 5.7 | 50.7 ± 1.5 | **11.9 ± 1.4** |
| | LA-SMT$^{AD}$ | 76.9 ± 4.7 | 57.6 ± 3.3 | <u>13.8 ± 2.0</u> |
| | LA-SMT$^{Lang}$ | 74.0 ± 7.2 | 55.5 ± 1.2 | <u>13.8 ± 2.0</u> |
| | | GUATEMALA | | |
| SOTA | `Gold LA` | – | – | 4.0 ± 0.5 |
| | `SAE` | <u>83.2 ± 2.5</u> | <u>73.4 ± 1.3</u> | 13.8 ± 4.6 |
| | `RetinaNet` | 75.4 ± 1.3 | 47.6 ± 8.8 | 26.2 ± 13.3 |
| | `YOLO` | **93.3 ± 0.4** | **91.4 ± 0.9** | 4.0 ± 0.5 |
| Other | TrOMR$^{LT}$ | 61.7 ± 3.7 | 54.1 ± 1.8 | 7.5 ± 1.0 |
| | TrOMR$^{AD}$ | 59.9 ± 3.5 | 49.6 ± 2.1 | 6.9 ± 0.8 |
| | TrOMR$^{Lang}$ | 60.3 ± 3.2 | 50.4 ± 2.3 | 6.9 ± 0.8 |
| | DAN$^{LT}$ | 70.3 ± 4.2 | 64.2 ± 1.4 | 4.0 ± 0.5 |
| | DAN$^{AD}$ | 68.3 ± 3.4 | 53.1 ± 1.1 | 4.3 ± 0.8 |
| | DAN$^{Lang}$ | 66.5 ± 3.4 | 66.5 ± 1.2 | 4.3 ± 0.9 |
| Ours | LA-SMT$^{LT}$ | 77.2 ± 3.9 | 67.9 ± 1.7 | **3.2 ± 0.8** |
| | LA-SMT$^{AD}$ | 77.3 ± 2.9 | 53.1 ± 1.1 | <u>3.8 ± 0.8</u> |
| | LA-SMT$^{Lang}$ | 75.8 ± 3.7 | 58.2 ± 2.4 | <u>3.8 ± 0.8</u> |

ness and effectiveness of our approach even in the absence of explicit layout supervision.

### 5.3 Quantitative analysis

The results presented in Tables 2 and 3 reveal significant insights about the behavior of the studied models. The LA-SMT method consistently delivers superior transcription results, demonstrating substantial improvements over the SOTA in challenging scenarios such as CAPITAN. This confirms that our end-to-end full-page transcription approach for historical music records is not only viable, but superior to the established methods for practical OMR applications. Implementing layout tokens during transcription is the most effective approach. This finding aligns with the outcomes reported in analogous fields [30, 31]. Notably, even without layout tokens, the model still outperforms existing state-of-the-art methods. These insights position the LA-SMT as a compelling candidate for state-of-the-art end-to-end full-page OMR, offering both improved performance and reduced training costs.

In the LA domain, the model demonstrates competitive performance, especially when utilizing layout tokens, which help the model implicitly learn document structure. However, the proposed ILA method falls short of the precision achieved by established object detectors such as YOLO. This performance gap is understandable, as LA-SMT generates bounding boxes from implicit representations rather than through the explicit training that specialized object detection models receive.

The proposed stopping criteria offer flexible alternatives for segmenting staves during transcription. However, their

**Fig. 3** Example of the snowball effect in the SEILS dataset. The image on the left corresponds to the output of the LA-SMT$^{LT}$, which achieves a 79% IoU and a 4.4% SER; the middle image shows the output fo YOLO, also with a 79% IoU and a 5.5% SER; and the right image presents the results from SAE, with a 70% IoU and a 7.22% SER. Green boxes indicate the ground-truth regions, while red boxes correspond to the predicted ones

effectiveness may depend on the characteristics of the input. In particular, the language-based criterion relies on the presence of clef symbols, which may not always be explicitly written in every staff, especially in handwritten or informal manuscripts, limiting its applicability. Moreover, while the inclusion of layout tokens improves performance, it requires additional annotation effort. Nevertheless, our proposal also supports a configuration without such tokens, avoiding any additional markup for staff segmentation. This allows experts to choose between higher accuracy and zero-cost annotation, depending on their available resources and the nature of their datasets.

Similar to what was observed in the printed document scenarios, a significant improvement in staff line detection is also evident in the case of handwritten manuscripts when compared to the other two transformer-based architectures, obtaining the best result with the lang criteria. However, both architectures are outperformed by our proposal, obtaining the same conclusions as in the printed scenario.

## 5.4 Qualitative analysis

The major insight from the analysis of Tables 2 and 3 is that, although LA-SMT is the best approach for end-to-end transcription, it underperforms state-of-the-art object detectors, such as the SAE and YOLO. Despite that, the LA-SMT remains a competitive approach for LA. In this section, we closely analyze the practical implications of these performance divergences and discuss the actual capabilities of the models when addressing full-page OMR.

### 5.4.1 Snowball effect

One of the main drawbacks of pipeline-based full-page OMR systems is the well-known *snowball effect*, where errors from the LA stage propagate to the single-staff transcription model. This effect is often overlooked in the literature, as it is assumed that a user intervenes manually between stages. As a result, reported outcomes tend to be overly optimistic. Figure 3 illustrates this phenomenon.

In this case, we observe that the performance of both YOLO and SAE degrades primarily due to detection inaccuracies. First, both methods fail to detect the final staff, which alone accounts for at least a 1% increase in SER. Furthermore, when comparing YOLO with SAE, we find that although SAE exhibits a 12% lower detection accuracy, this results in a substantial 32% drop in transcription performance. These results highlight a key weakness of layout-based pipelines: despite being highly accurate, the transcription stage remains sensitive to minor misalignments or imperfections in the extracted regions. In contrast, the LA-SMT avoids cascading such errors by learning to read the full score content explicitly, while implicitly inferring its layout.

### 5.4.2 Language model support

The LA-SMT is essentially a text generator conditioned on the visual content of the image to transcribe. This conditioning allows the model—through its integrated language modeling ability—to capture the internal structure of music documents. This characteristic is particularly beneficial, as it encourages the generation of syntactically coherent transcriptions, even when the visual input is partially ambiguous or imprecise. This stands in contrast to staff-wise transcription approaches based on CTC, which rely heavily on a strong alignment between the visual features and the predicted sequence, making them more susceptible to misalignments or detection errors.

Although our pipeline proposal was pre-trained with a synthetic corpus, this step was not intended for vocabulary memorization, but rather to help the model learn general representations of complex reading orders in full-page scores—a challenge that is particularly difficult for transformer-based models. This design choice follows the recommendations of

**Fig. 4** Language support example in the CAPITAN dataset. The image on the left shows the output of the LA-SMT$^{LT}$, which achieves 68% IoU and 8.39% SER, while the image on the right corresponds to YOLO, reporting 92% IoU and 10.32% SER. Green boxes indicate the ground truth, while red boxes denote the predicted areas

**Fig. 5** Example of the actual staff regions detected by the method on a test sample from the GUATEMALA dataset. The image on the left shows the output of LA-SMT$^{LT}$, which reports 77% IoU and 0% SER, while the image on the right corresponds to YOLO, with a 95% IoU and a 1.32% SER. Green boxes indicate the ground truth, and red boxes represent the predicted regions

prior work [10], which reported that such pre-training is necessary.

Figure 4 presents an example from the CAPITAN dataset, which represents the most challenging source evaluated. In this case, the LA-SMT performs poorly in terms of staff detection, while YOLO achieves a remarkably high bounding box accuracy of 92%. Despite this, our proposed approach achieves 23% better transcription performance compared to the detection-based pipeline. This advantage stems from the internal language modeling capabilities of the Transformer, which enable it to produce consistent and syntactically coherent outputs even in visually ambiguous scenarios, where the correlation between image content and transcription may be weak. Notably, while the regions predicted by the LA-SMT may appear suboptimal when compared to the human-annotated ground truth, they still capture sufficient contextual information for accurate transcription and staff retrieval. These results demonstrate that LA-SMT offers greater robustness as a transcription model in challenging scenarios, even when layout detection does not strictly align with human expectations.

### 5.4.3 What the LA-SMT actually detects

One of the main concerns emerging from the analysis in Section 5 is that, although the LA-SMT has proven to be a superior transcription approach, it does not match the performance of state-of-the-art detection pipelines in terms of layout accuracy. However, at no point we examined what the LA-SMT is actually extracting as the bounding box for each staff. This is clearly illustrated in Figures 3 and 5

After analyzing the examples, we observe that the apparent lower IoU scores of our approach compared to YOLO stem primarily from a fundamental difference in region identification strategy, rather than from inferior detection performance. The ILA method, which derives bounding boxes from attention maps during transcription, prioritizes regions that contain actual musical content, deliberately excluding empty staff areas. This contrasts with the corpus annotations, where annotators—favoring simplicity and alignment with conventional detection training—included the entire staff region, even when parts of it contained no relevant information. However, it is important to note that, in practice, many available corpora are already annotated following criteria defined externally by annotators, often without documentation or standardization. These annotations are usually shaped by subjective choices such as annotators' expertise, habits, or task-specific priorities. As such, the suitability of these ground truths for evaluating our approach may vary significantly. Since these factors are external to our model and outside the scope of this work, we cannot assume full control over how ideal or consistent these annotations may be for our method's specific architecture in real-world scenarios.

We found that this issue occurs predominantly in the handwritten datasets. Specifically, over 72% of the images in GUATEMALA and 15% in CAPITAN are affected, indicating that conventional evaluation protocols may significantly underestimate the actual effectiveness of our method. We manually revised both datasets and repeated the experiments using the corrected ground truth. The results obtained from these updated corpora are reported in Table 4.

The results show improvements in both IoU and mAP. Specifically, the GUATEMALA benchmark improves over 3.2% and 7.8% in both metrics, setting the LA-SMT very close to SAE. It also improves 6.8% and 2.7% in CAPITAN, where this annotation issue happens less frequently. This shows that, in practice, the content-focused strategy from

**Table 4** Average IoU and mAP between the layout analysis results retrieved by LA-SMT$^{LT}$ in the handwritten datasets, where most layout inconsistencies have been detected

| Dataset | Avg. IoU | Avg. mAP |
|---|---|---|
| *Guatemala* | | |
| Not corrected | 77.2 | 67.9 |
| Corrected | 80.3 | 73.2 |
| *Capitan* | | |
| Not corrected | 70.5 | 50.7 |
| Corrected | 75.3 | 52.1 |

the LA-SMT proves equally effective for transcription, as it selectively omits empty regions that do not contribute meaningful information. In this context, our evaluation should be considered a conservative estimate of the actual detection capabilities of the model, especially in the handwritten sets, where the divergences in labeling do significantly affect the performance reports.

Therefore, the LA-SMT demonstrates to be very competent as a state-of-the-art method for full-page end-to-end OMR, excelling in transcription while offering strong staff-region detection. This content-centric approach represents a substantial advancement over existing pipelines, despite what may appear as a numerical disadvantage under standard object detection metrics.

## 6 Conclusion

This paper presents the LA-SMT, an end-to-end full-page OMR system that simultaneously transcribes the content of an input music score and implicitly identifies the regions in which that content is located—essential for practical applications. The model integrates autoregressive transcription with Transformers with an ILA algorithm that leverages the attention mechanism of the model to extract regions of interest.

We assess the performance of three variants of our approach against pipeline-based OMR methods—currently the standard in the literature—across four well-established historical datasets, both printed and handwritten. The results reveal that our approach outperforms all baselines in transcription accuracy. We also identify an important trade-off in region extraction: while traditional object detection metrics favor state-of-the-art models, they often fail to reflect their practical impact. Even bounding boxes with slightly lower detection scores can still capture all relevant musical content, significantly reducing transcription errors. Since our method derives bounding boxes directly from the attention matrices, it reliably includes all graphical symbols. As a result, our approach is not only strong in transcription, but also highly competitive in region detection, establishing a new state-of-the-art for end-to-end OMR.

This work highlights promising avenues for future research. One improvement could be to refine the attention mechanisms of LA-SMT through music-notation-aware learning, conditioning the attention matrices to be more accurate. The data-intensive nature of the model could also be improved through optimization strategies and self-supervised learning techniques, rely on large annotated datasets while improving overall performance.

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**Availability of data and materials:** No datasets were generated or analysed during the current study.

**Code Availability** The source code of the model will be available after review at: https://github.com/antoniorv6/smt-implicit-la.

## Declarations

**Competing interests** The authors declare no competing interests.



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